LEARNING SESSION S11

Automation on Data

Automate the boring work with Excel, SQL, Python, Power BI, Tableau and ML/AI, with a complete weekly-report example

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

Contents

Why & how

Excel

SQL

Python

Power BI & Tableau

ML & AI

End-to-end example

Resources

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⚙️ Automation on Data

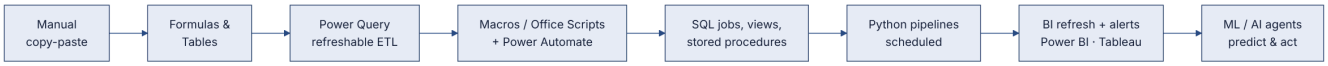
Automate the boring work with Excel, SQL, Python, Power BI, Tableau and ML/AI, with a complete weekly-report example

Additional chapter · from manual copy-paste to scheduled, self-refreshing pipelines

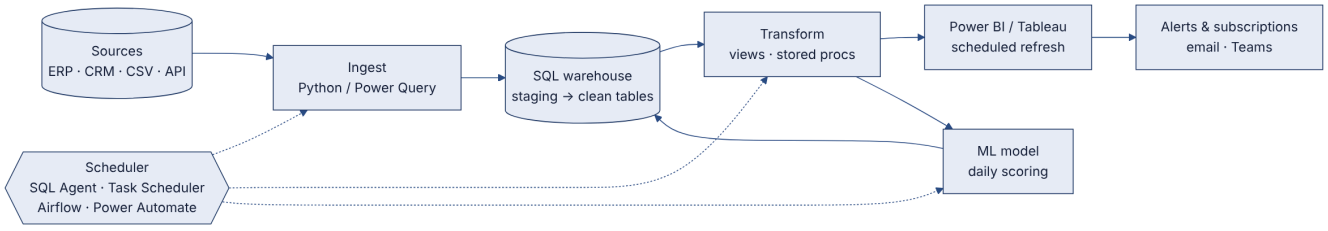
Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s11**

Why & how

Automation turns a report you rebuild every week into one that rebuilds itself. It saves hours, removes copy-paste errors, and frees you to analyse instead of assemble.



The automation ladder: climb one rung at a time.



A typical automated analytics pipeline with a scheduler driving each step.

What to automate first

Signal	Example
Repeated on a schedule	Weekly MIS, month-end reconciliation
Same steps every time	Download → clean → pivot → paste into PPT
Error-prone	Manual VLOOKUPs across 10 files
Time-consuming	More than 30 minutes per run

Tip ROI check: (minutes saved per run × runs per year) vs build time. A 2-hour weekly task automated in 8 hours pays back in a month.

Excel

Technique	Automates	How
Tables + formulas	Calculations on new rows	Ctrl+T; structured references
Power Query	Import, clean, combine	Get Data → From Folder; Refresh All; set "Refresh on open"
Pivot auto-refresh	Summaries	PivotTable Options → Refresh data when opening the file; auto-refresh (365)
Macros / VBA	Formatting, exporting PDFs, emailing	Record a macro; edit in Alt+F11
Office Scripts + Power Automate	Cloud runs on a schedule or trigger	Automate tab → Record; Power Automate "Run script" on a recurrence
Copilot Agent Mode	Multi-step builds from a prompt	"Refresh, update the summary and highlight anomalies"

```
' VBA: save every sheet as a separate PDF
Sub ExportSheetsToPDF()
    Dim ws As Worksheet
    For Each ws In ThisWorkbook.Worksheets
        ws.ExportAsFixedFormat Type:=xlTypePDF, _
            Filename:=ThisWorkbook.Path & "\" & ws.Name & ".pdf"
    Next ws
End Sub
```

SQL

```
-- 1. A view: the cleaned, reusable dataset every report reads
CREATE VIEW rpt.vw_weekly_sales AS
SELECT DATETRUNC(WEEK, o.order_date) AS week_start, c.city, p.category,
       SUM(oi.qty * oi.price) AS revenue, COUNT(DISTINCT o.order_id) AS orders
FROM Orders o JOIN Customer c ON o.customer_id = c.customer_id
JOIN Order_Item oi ON o.order_id = oi.order_id JOIN Product p ON oi.product_id = p.product_id
GROUP BY DATETRUNC(WEEK, o.order_date), c.city, p.category;

-- 2. A stored procedure that loads a summary table
CREATE PROCEDURE rpt.usp_load_weekly AS
BEGIN
    TRUNCATE TABLE rpt.weekly_sales;
    INSERT INTO rpt.weekly_sales SELECT * FROM rpt.vw_weekly_sales;
END;
-- 3. Schedule it: SQL Server Agent → Jobs → New Job → Step: EXEC rpt.usp_load_weekly →
Schedule: Mondays 06:00
--    (Express edition has no Agent: use Windows Task Scheduler with sqlcmd, or Azure Data
Factory / Fabric pipelines)
```

Also: triggers (react to inserts/updates, e.g. audit logs), `MERGE` for incremental upserts, Database Mail for alert emails.

Python

```python
# weekly_report.py — pull data, build Excel report, email it
import pandas as pd, sqlalchemy as sa, smtplib, datetime as dt
from email.message import EmailMessage

engine = sa.create_engine("mssql+pyodbc://@localhost/RetailDB?
driver=ODBC+Driver+18+for+SQL+Server&trusted_connection=yes&TrustServerCertificate=yes")
df = pd.read_sql("SELECT * FROM rpt.weekly_sales WHERE week_start >= DATEADD(WEEK, -8,
GETDATE())", engine)

summary = df.pivot_table(index="city", columns="week_start", values="revenue",
aggfunc="sum").round(0)
out = f"weekly_sales_{dt.date.today():%Y%m%d}.xlsx"
with pd.ExcelWriter(out) as xw:
    summary.to_excel(xw, sheet_name="Summary")
    df.to_excel(xw, sheet_name="Data", index=False)

msg = EmailMessage(); msg["Subject"] = "Weekly sales report"; msg["From"] =
"reports@company.com"; msg["To"] = "team@company.com"
msg.set_content("Attached: last 8 weeks by city.")
msg.add_attachment(open(out, "rb").read(), maintype="application", subtype="octet-stream",
filename=out)
with smtplib.SMTP("smtp.company.com", 587) as s:
    s.starttls(); s.login("reports@company.com", "USE_A_SECRET_STORE"); s.send_message(msg)
```

Schedule with	How
Windows Task Scheduler	Create Basic Task → Weekly → Start a program: `python C:\jobs\weekly_report.py`
cron (Linux/Mac)	`0 7 * * 1 /usr/bin/python3 /jobs/weekly_report.py` (Mondays 07:00)
Cloud	GitHub Actions schedule, Azure Functions, AWS Lambda, Fabric notebooks
Orchestrators	Apache Airflow, Prefect, Dagster for multi-step pipelines with retries and monitoring

> Never hard-code passwords in scripts; use environment variables or a secrets manager.

Power BI & Tableau

Power BI

- **Scheduled refresh** of semantic models (gateway for on-premises data); **incremental refresh** for large tables.
- **Subscriptions**: email report snapshots on a schedule; **data alerts** on dashboard tiles.

- **Power Automate**: trigger flows from an alert, a button in the report, or after refresh ("refresh dataset then post to Teams").
- **Fabric**: Data Factory pipelines and notebooks feed Lakehouse tables; Direct Lake reports update without import refresh.
- **Copilot** summaries for automated narrative.

Tableau

- **Extract refresh schedules** on Tableau Cloud/Server.
- **Tableau Prep** flows scheduled with Prep Conductor for repeatable cleaning.
- **Subscriptions** and **data-driven alerts**.
- **Tableau Pulse** pushes metric changes with AI-written insights to email or Slack.

ML & AI

Automation	Example	Tools
Batch scoring	Score all customers for churn every Sunday; write scores to SQL for the CRM and Power BI	Python script + scheduler; Azure ML / Fabric
Retraining triggers	Retrain when accuracy drops or data drifts	MLflow, Azure ML pipelines
Anomaly alerts	Flag days when sales fall outside expected range	Prophet/ETS forecast + rule; Power BI anomaly detection
LLM text classification	Tag 10,000 complaints by theme nightly	OpenAI/Anthropic API, Excel `=COPILOT()` for small sets
AI agents	"Every Monday: query sales, update dashboard, write commentary, ask for approval, email"	Copilot Studio, ChatGPT/Claude agents, n8n/Power Automate with AI steps

```
# Nightly: classify new complaints with an LLM (pattern; use your company-approved provider)
import pandas as pd
from anthropic import Anthropic          # or the OpenAI / Azure OpenAI SDK
client = Anthropic()                     # API key from environment variable
themes = ["Delivery delay", "Refund", "Product quality", "App issue", "Other"]
def classify(text):
    r = client.messages.create(model="claude-haiku-4-5-20251001", max_tokens=10,
        messages=[{"role": "user", "content": f"Classify into one of {themes}. Reply with the
label only.\n\n{text}"}])
    return r.content[0].text.strip()
new = pd.read_csv("complaints_today.csv")
new["theme"] = new["complaint_text"].map(classify)
new.to_csv("complaints_tagged.csv", index=False)
```

> **Tip** Validate AI labels on a hand-checked sample (say 200 rows) before trusting them, and re-check monthly.

End-to-end example

Weekly sales review, fully automated

1. **06:00 Mon** – SQL Agent runs `usp_load_weekly`: cleaned weekly table updated.
2. **06:15** – Python scores churn risk for active customers and writes scores back to SQL.
3. **06:30** – Power BI scheduled refresh; Copilot summary updated.
4. **06:45** – Power Automate checks KPIs; if revenue is down more than 10% week on week, posts an alert in Teams with the top 3 contributing cities.
5. **07:00** – Subscription emails the report PDF to managers; analyst reviews exceptions only.

Before: 4 hours every Monday of downloads, VLOOKUPs and copy-paste. **After:** 15 minutes reviewing exceptions.

Resources

- ▶ YouTube: Simplilearn Excel VBA tutorial · ▶ YouTube: Simplilearn Power Automate tutorial · ▶ YouTube: Simplilearn Python automation
- ▶ YouTube: Leila Gharani Office Scripts Power Automate Excel · ▶ YouTube: SQL Server Agent job schedule tutorial
- ▶ YouTube: Power BI scheduled refresh gateway tutorial · ▶ YouTube: Tableau Prep Conductor schedule flow
- 📘 Automate the Boring Stuff with Python · 📘 Power Automate documentation
- 📘 Office Scripts in Excel · 📘 Apache Airflow